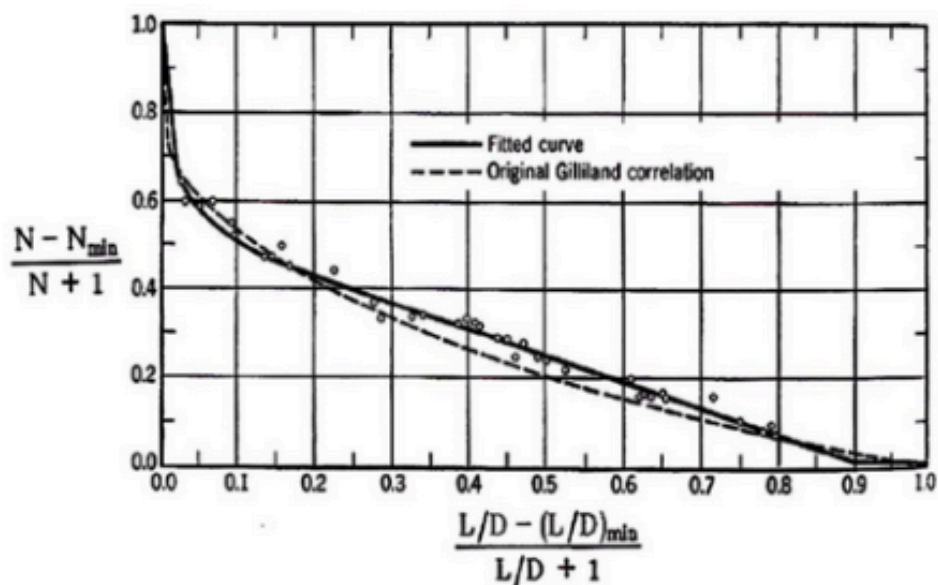
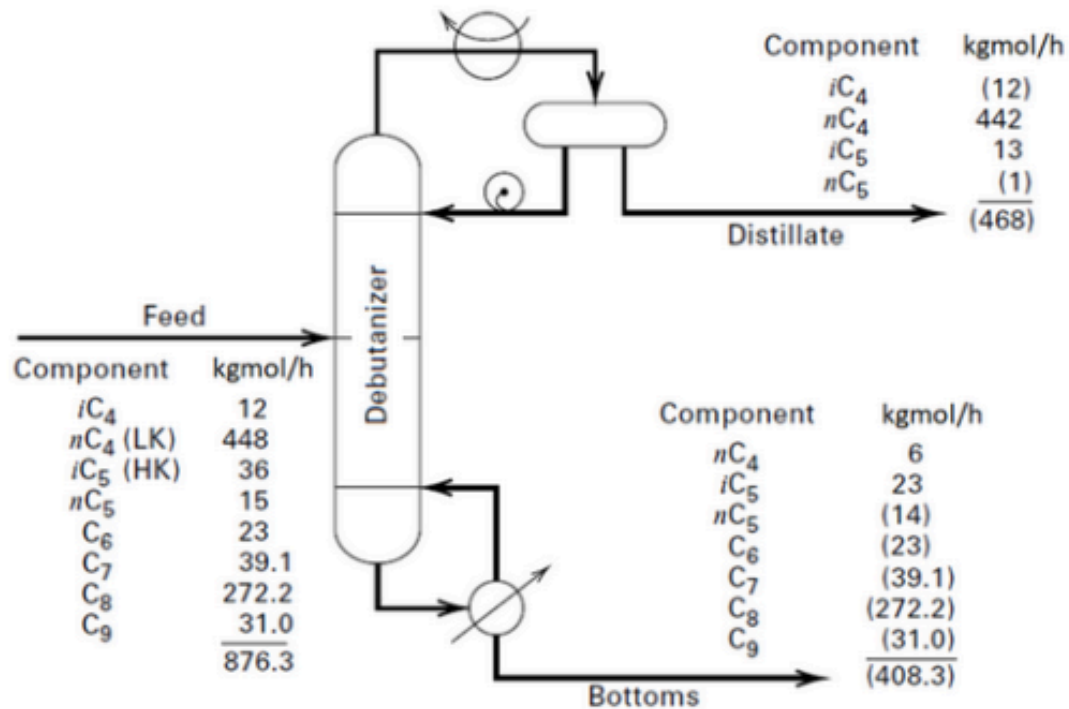


CH620003
PROCESS MODELLING AND
SIMULATION
ASSIGNMENT-2

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Q1: FUG TECHNIQUE

- (a) For the debutanizer shown here, estimate the N_{min} by the Fenske equation. Assume a uniform operating pressure of 80 psia (552 kPa) throughout and utilize the ideal K-values.
- (b) Estimate the product distributions for nonkey components by the Fenske equation.
- (c) Calculate the minimum reflux, based on Underwood equation.
- (d) Use the Gilliland chart (included below) to estimate the number equilibrium-stage required for the debutanizer, for a reflux 30% greater than the minimum reflux.



Ans: Used Fenske, Underwood and Gilliland Correlation to get the following result from the python simulation which is there in the q1.py file.

Results:

Estimated relative volatility (alpha): **0.48**

Minimum number of stages (**N_{min}**): **7.00**

Nonkey component distributions:

C9: Distillate = 0, Bottoms = 31.0

C6: Distillate = 0, Bottoms = 23

C7: Distillate = 0, Bottoms = 39.1

iC4: Distillate = 12, Bottoms = 0

C8: Distillate = 0, Bottoms = 272.2

nC5: Distillate = 1, Bottoms = 14

Minimum reflux ratio (**R_{min}**): **0.16**

Actual number of stages required: **22.00**

For the Gilliland Chart used the following correlation as we were supposed to use a functional form to incorporate that in the simulation code.

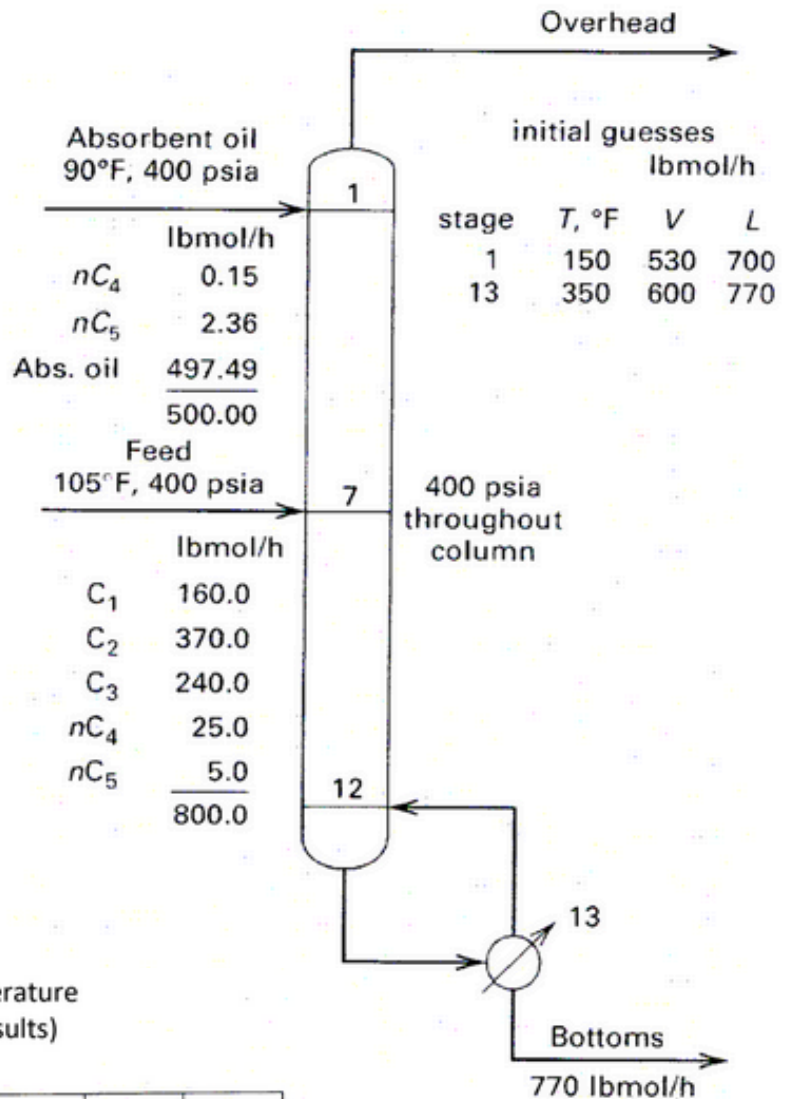
$$Y = 1 - \exp \left[\frac{(1 + 54.4X)(X - 1)}{(11 + 117.2X)\sqrt{X}} \right] \quad (6)$$

$$Y = \frac{N - N_{\min}}{N + 1} \quad (7)$$

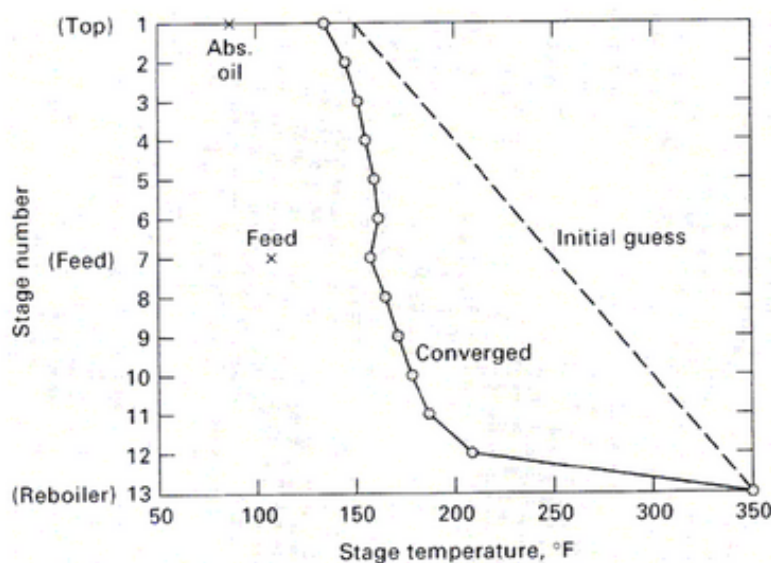
$$X = \frac{R - R_{\min}}{R + 1} \quad (8)$$

Q2: SIMULTANEOUS – CORRECTION (SC) METHOD

A reboiled absorber is used for the separation of a hydrocarbon vapour feed. Here the absorbent oil in *n*-decane. The 770 lbmol/h of bottom feed corresponds to the amount of C_3 and heavier in the two feeds, so the column is designed as a deethanizer. Calculate the stage temperatures, interstage vapour and liquid component flow rates, and reboiler duty using the SC method.

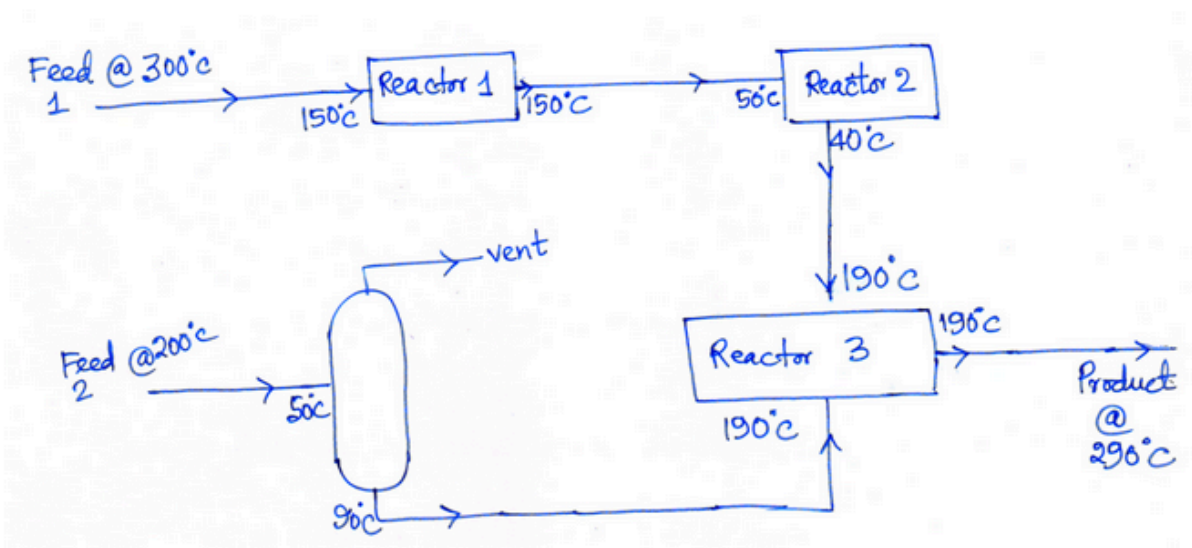


Sample solution for the stage temperature
(for validating your calculations / results)



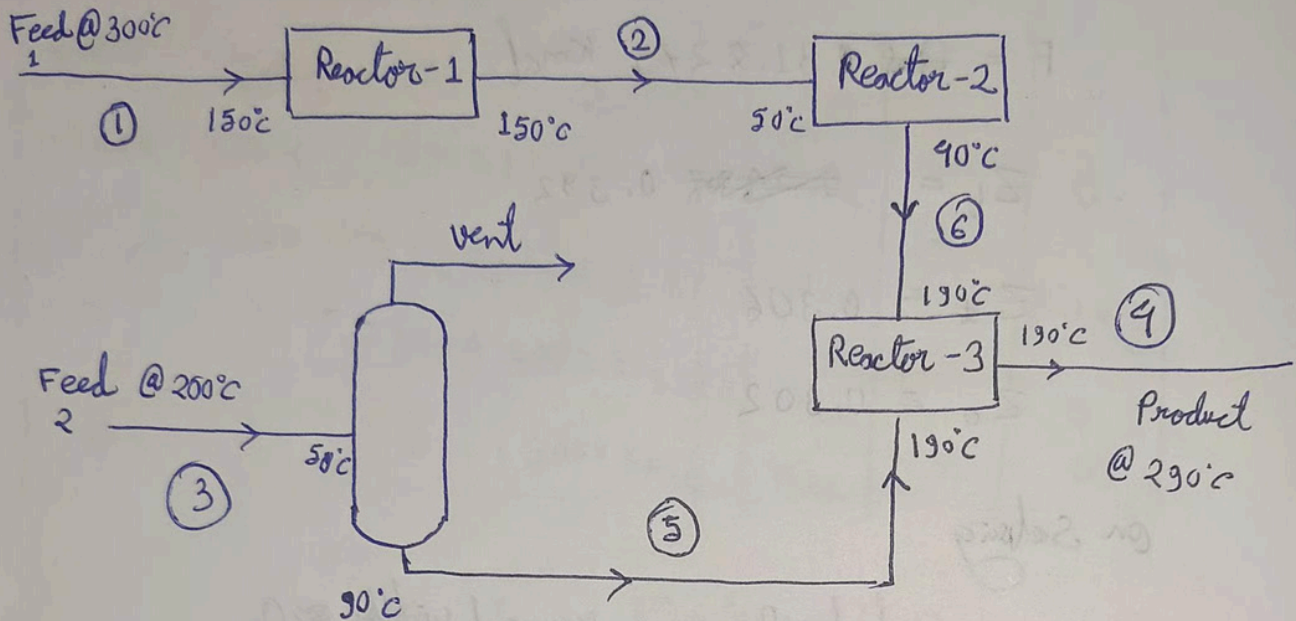
Q3: PINCH ANALYSIS AND HEAT EXCHANGER NETWORK INTEGRATION

- Prepare the composite temperature-enthalpy curve for $\Delta T_{min} = 10^\circ\text{C}$. Estimate the minimal heating and cooling loads, and identify the pinch temperature.
- Estimate the pinch temperature from the temperature cascade procedure and check with the result from the T-H diagram.
- Design an optimized heat exchanger network connection that requires minimal hot and cold utility energies.
- Prepare a new process flowsheet integrating the heat exchangers in the process flow diagram.



Stream No.	Flowrate, $\dot{m}$ (kg/s)	C_p (kJ/kg°C)	T_{in} (°C)	T_{out} (°C)
1	10.00	0.8	300	150
2	2.50	0.8	150	50
3	3.00	1.0	200	50
4	6.25	0.8	190	290
5	10.00	0.8	90	190
6	4.00	1.0	40	190

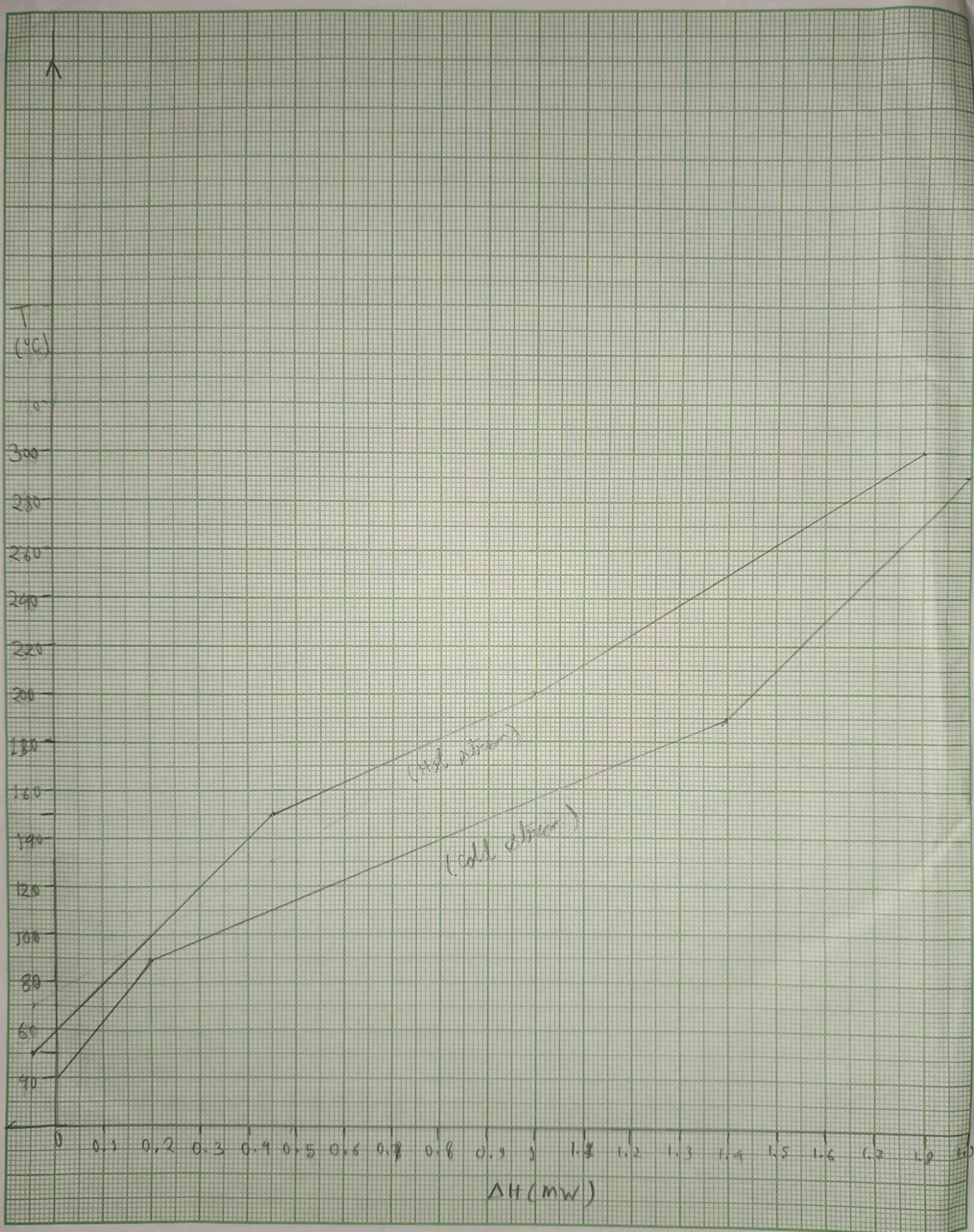
Q.3. given the following diagram;



Stream no.	Flow rate $\dot{m}$ (kg/s)	C_p (kJ/kg $^{\circ}$ C)	T_{in} ($^{\circ}$ C)	T_{out} ($^{\circ}$ C)	$\dot{m} C_p$ (kJ/s $^{\circ}$ C)	ΔH (kJ/s) (kW)
1	10	0.8	300	150	8	-1200
2	2.5	0.8	150	50	2	-200
3	3	1	200	50	3	-450
4	6.25	0.8	190	290	5	+500
5	10	0.8	90	190	8	+800
6	4	1	40	190	4	+600

$$\Delta H_{re} = -1850 \text{ kW}$$

$$\Delta H_{pre} = +1900 \text{ kW}$$



for hot streams:

$T_{range} (^{\circ}C)$	$\dot{m} C_p$	number of stream	$\Delta H (kW)$
50 - 150	5	2	-500 = -0.5 MW
150 - 200	11	2	-550 = -0.55 MW
200 - 300	8	1	-800 = -0.8 MW

for cold streams:

$T_{range} (^{\circ}C)$	$\dot{m} C_p$	number of stream	$\Delta H (kW)$
40 - 90	4	1	+200 = 0.2 MW
90 - 190	12	2	+1200 = 1.2 MW
190 - 290	5	1	+500 = 0.5 MW

now, in the graph we can observe that the ΔT_{min} is at $\Delta H = 0.2$ MW,

now the pinch temperature for the cold stream will be

90°C

and that for the hot stream will be 100°C,

and Total Heat exchanger load will be = 1.8 MW.

and, unutilized ~~heat~~ Heat load = 0.05 MW

cooling load = 0.1 MW.

(b)

now,

$$T_c^* = T_c + \frac{\Delta T_{\min}}{2}$$

$$T_h^* = T_h - \frac{\Delta T_{\min}}{2}$$

Stream
no.

Type

 T_s T_t T_s^* T_t^*

1

Hot

300

150

295

145

2

Hot

150

50

145

45

3

Hot

200

50

195

45

4

Cold

190

290

195

295

5

Cold

90

190

95

195

6

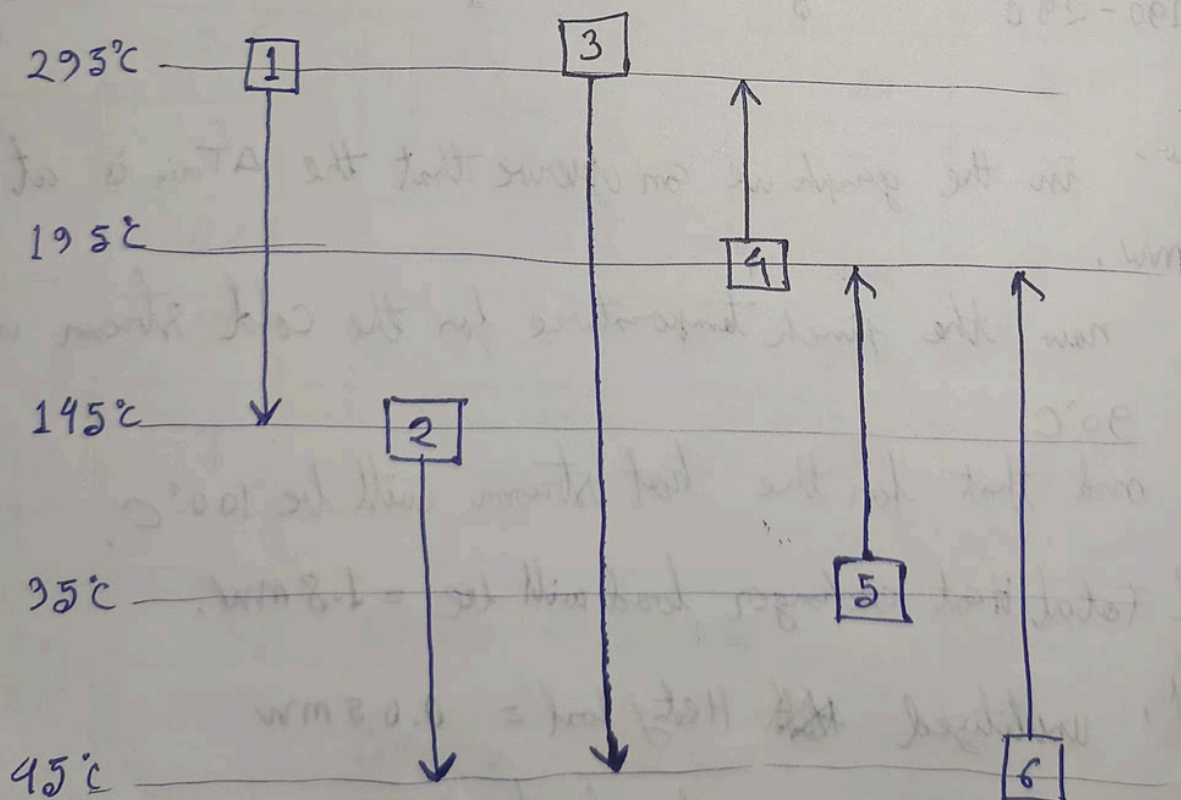
Cold

40

190

45

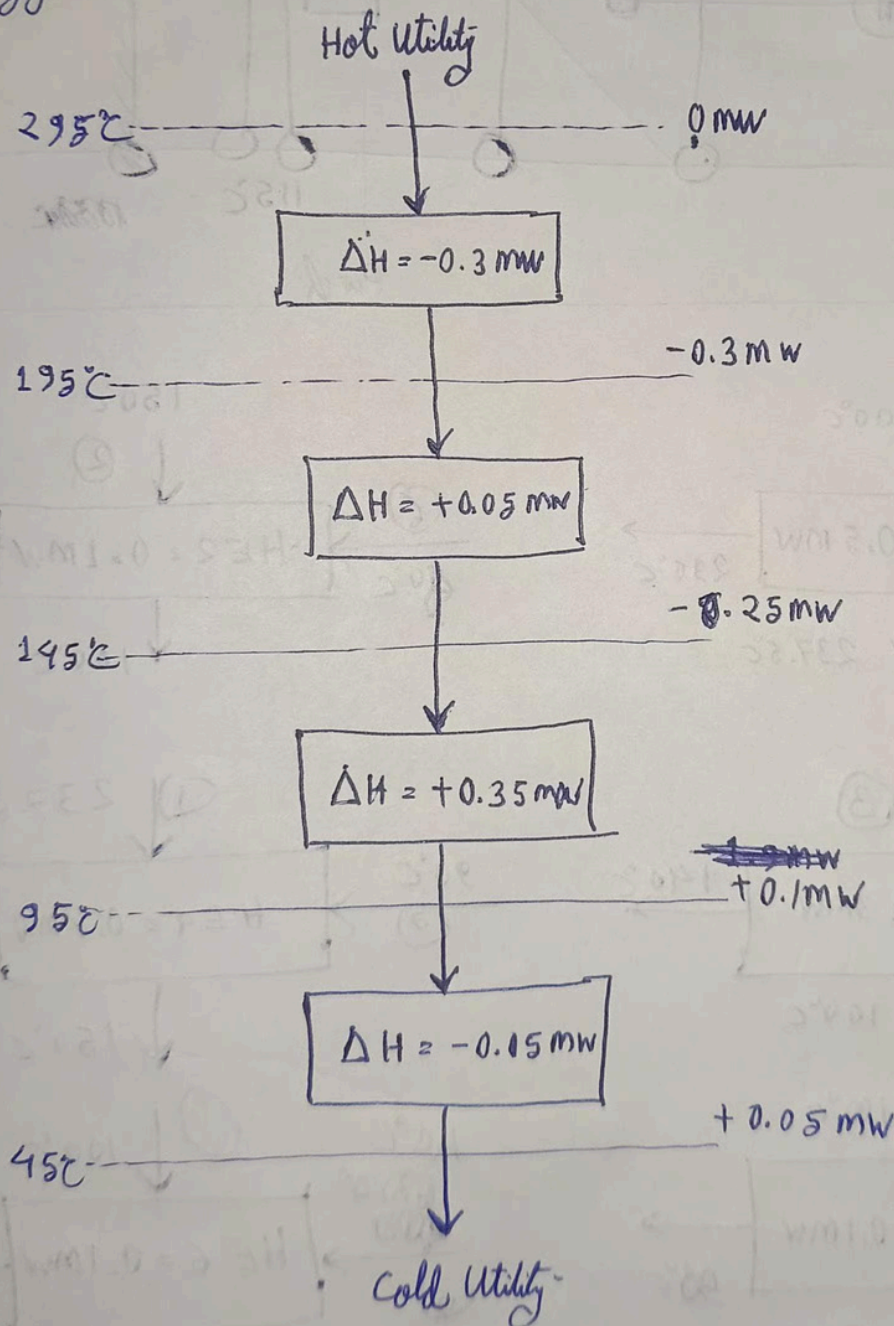
195



Heat Balances:

$\Delta T(^{\circ}\text{C})$	$\sum C_{P_c} - \sum C_{P_H}$	$\Delta H \text{ (MW)}$	Surplus/Deficit
100	-3×10^{-3}	-0.3	Surplus
50	$+1 \times 10^{-3}$	+0.05	Deficit
50	$+7 \times 10^{-3}$	+0.35	Deficit
30	-1×10^{-3}	-0.05	Surplus

cascade Energy from



(C) ∞

